FOR EVERYONE — THE RESEARCHER, THE JOURNALIST, THE ONE WHO CHECKS

THE ABSENCE AUDIT — THE METHOD, AT NATIONAL SCALE

Tunisia, audited — the complete audit

Five absences. Nine legs. One metabolism. Every number recomputed in code, every correction applied into the design, every claim graded.

Provenance law: MEASURED = cited anchor · DERIVED = arithmetic on measured values · ASSUMED = declared default. commercially-validated = 0 — no buyer has signed anything; published as zero on purpose. Generated 2026-09-09 by nations_audit.py / make_nations_pdfs.py — the same code that renders the page.

1. The five absences and the one metabolism

Five structural absences — monetary sovereignty (debt $40.5B, ~79% of GDP, 26.2% of export revenues to service), sovereign compute (InstaDeep: founded in Tunis, sold for $680M), strategic materials (phosphogypsum at Gabès costing the sea >$58M/yr), water, nutrition — answered by nine cash-positive legs on one metabolism of power, water, data and workforce. The audit is finished; the corrections are in the design.

2. The FX bridge — every dollar, by leg

Each leg below is volume × price × utilization, with the double-counting checks applied — the design carries the checked ledger.

LEG	TYPE	VOLUME	PRICE	$M/YR	GRADE
Strategic metals (U + heavy REE, acid stream)	gross export	716,500 lb U_3O_8 + 585 t REO	$80-95/lb U_3O_8 · $30/kg REO	$75-86M	**STRONG**
Solar fuel substitution (500 MW)	import substitution	745 GWh/yr displaced (876 GWh less 131 GWh to H_2)	$142/MWh (gas $12/MMBtu x 11 MMBtu/MWh + $10/MWh O&M); government anchor: 1,100 GWh ≈ $125M/yr gas saving for the same 500 MW fleet (Pg14)	$95-116M	**PLAUSIBLE**
Hydrogen → ammonia (loop-internal offtake)	import substitution	2.63 kt H_2/yr → 14.9 kt NH_3 for the GCT chain	$500/t imported ammonia displaced (external offtake upside: $8-9.5/kg H_2)	$7-25M	*SPECULATIVE*
Terfas (desert truffles)	gross export	940 t/yr mid (2,500 ha x 376 kg/ha; 70% exported)	EUR20-40/kg farm-gate	$14-29M	**PLAUSIBLE**
Food import substitution	import substitution	1% of the $3.0B food import bill	imported protein $15-25/kg vs local $5-10/kg	$30-30M	*SPECULATIVE*

LEG	TYPE	VOLUME	PRICE	$M/YR	GRADE
		(frame; volume bridge is a pilot deliverable)			
Compute colocation services	gross export (services)	colo park, tenant-dependent	market colocation rates	$15-15M	*SPECULATIVE*
ESG financing delta	financing delta (not FX)	100-150bp on ~$800M portfolio debt	interest saved, not foreign exchange	$8-12M	**PLAUSIBLE**
Total external-account effect (mid)				**~$279M/yr**	**$80M strong · $137M plausible · $61M speculative · $0 verified**

By kind: exports $117M · import substitution $152M · financing delta $10M (interest saved, honestly labeled as not-FX). **By grade:** $80M strong · $137M plausible · $61M speculative · $0 verified — nothing is commercially validated yet, and that ledger reads zero on purpose.

3. The energy ledger — every kWh, by leg

LEG	GWH/YR	THE FORMULA
Pump-as-turbines (return flows)	3.9	70% × 150k m^3/d × 50 m × 75% × 8,760 h
Heat (kiln + exotherm)	30	9% of 333 GWh envelope
Flexibility (load shift)	20	dispatch value — not recovered joules
CO&sub2; mineralized	30 kt	mass, not energy
Total	**33.9 recovered + 20 dispatch**	**isobaric ERD inside the 4 kWh/m^3 benchmark — never double-counted**

4. Phase 0 — every dollar, itemized

ITEM	COST	WHAT IT BUYS
Water panel: 100 points, full suite (F, Cd, Pb, Cr, As, salinity, nitrate)	$35,000	the first anchor measurement
Acid-stream assays: 12 points (U, REE, Ra-226, Po-210, full metals) + GCT access	$40,000	the second anchor measurement
Emissions inventory: 30 passive samplers (PM2.5/SO2/fluorides) + lab	$25,000	baseline for the health scoreboard
Cohort protocol: 3,000-family registry, ethics, data platform v0	$35,000	the DALY ledger becomes MEASURED

ITEM	COST	WHAT IT BUYS
Health pilot first tranche: 100 households + 2 schools, 90 days	$58,000	filters, clean-air boxes, KPIs
Legal + finance: health fund, industrial SPV, land-lease pre-agreements, tender registration	$25,000	structure before scale
Engineering + project management	$35,000	the 90 days are run
List price	**$253,000**	**list price $253k; the two anchor measurements cost $75k and land first — the claimed range is the phased-execution plan, not the list**

5. Every claim, graded

CLAIM	GRADE	WHY
$40.5B debt (~79% GDP); 26.2% of export revenues to service	**STRONG**	MEASURED via World Bank API on 2026-09-09: debt $40.46B, debt/GDP 78.7%, service 26.17% of export revenues (2024)
$3.0B food and $4.9B fuel imports	**STRONG**	fuel $4.895B verified on WITS/Comtrade (HS27, 2024); food per the WB food classification
InstaDeep founded in Tunis, sold for $680M	**STRONG**	public deal record; the emblem for talent without infrastructure
Gabès: >$58M/yr of REE value lost to the sea	**STRONG**	Science of the Total Environment (2021): up to 1,523.67 t/yr REE, ~$58M/yr losses
U + heavy REE from the acid stream, ~$80M/yr	**STRONG**	flowsheet industrial since 1978 (Freeport), patent expired, Morocco building it now; volume derived from cited primitives; Gafsa-specific assays are Phase 0
~$285M/yr external-account effect, tiered	TIERED	$80M strong (proven flowsheet/cited anchor) · $137M plausible (gated) · $68M speculative (assumed volume/price)
33.9 GWh/yr recovered + 20 GWh/yr dispatch value	DERIVED	every leg computed from cited primitives; dispatch value honestly labeled — not joules
$2,592 per healthy year (water rung)	**PLAUSIBLE**	arithmetic DERIVED in code; DALY anchors transferred from Iranian studies — the cohort baseline makes it MEASURED
Phase 0 = $150-250k, itemized	**PLAUSIBLE**	itemized list price $253k; the two anchor measurements cost $75k and land first
Commercially validated	**ZERO**	no buyer has signed anything — published as zero on purpose

6. The hidden dependencies

DEPENDENCY	TYPE	STATUS	NOTE
Desal brine permit	regulatory	in design	Gabes outfall / diffuser; permit required before SWRO FID

DEPENDENCY	TYPE	STATUS	NOTE
PG chemistry (Ra-226 partitioning)	technical	Phase 0 assay	flowsheet keeps radium in gypsum (Freeport precedent); re-verify on Gafsa stream
Uranium handling	regulatory	gate	CNSTN licensing, NORM transport, export controls
REE separation capex	capital	later phase	+$244M solvent-extraction plant beyond the ~$100M circuit
Solar bankability	financial	gate	DSCR ~0.9-1.0 at the record-low tariff; bankable region or grant support required
Grid connection	infrastructure	later phase	3 GW connection + 400 kV upgrades ~$450M
Compute accelerators	regulatory	gate	export-control licenses; colocation until an anchor tenant is licensed
Truffle yield	agronomic	gate	376 kg/ha is the 20-yr Murcia average; gate = 3-season record >=200 kg/ha
Export prices	market	exposure	U spot $80-95/lb; truffle EUR20-40/kg — no contracted offtake yet
Financing	financial	uncommitted	DFI channel modeled for solar/desal; nothing committed

7. The structure and the pilot/capital separation

The design is six independently bankable modules sharing infrastructure. The loop is the outcome, never the precondition — no module's economics depends on another module succeeding, and every gate is public. World Bank CCDR (2023): Tunisia's water-sector investment need ~$17.1B and energy-sector ~$34.7B through 2050. The page is not that program and does not claim to be: Phase 0 is $150-250k of measurements, and the whole core build is ~$250-650M phased over years 2-5, module by module, each gated. The CCDR numbers confirm the scale of the problem; they do not compete with a pilot gate.

Africa Energy Portal (2026): 500 MW of utility-scale solar approved (~$400M investment). Same class as the page's P3 module (500 MW) — the record-low ~$31/MWh awards are the reference the DSCR gate is tested against.

8. The Fed transmission — measured, not restated

World Bank API, measured 2026-09-09 (Pg1): external debt **$40.46B**, **78.7% of GDP**; debt service **26.2% of export revenues**; reserves **$9.34B** = 4.3 months of imports; food $3.00B + fuel $4.90B of imports; CA -1.5% of GDP; growth 2.5%, inflation 5.2% (2025).

Channel 1 — the debt trap: the "original sin" (Pg12) — no long-term borrowing in the dinar abroad — exposes the state to the Fed cycle (Pg13): Fed hikes lift global yields, refinancing turns expensive, and the 26.2% of export revenues that debt service absorbs compounds while reserves (4.3 months) drain.

Channel 2 — imported inflation: the rate differential pulls capital to US safe assets, the dinar depreciates, and the dollar invoice on food ($3.00B) and fuel ($4.90B) inflates. The BCT — autonomous, non-monetizing under Organic Law 2016-35 (Pg4/Pg5) — must hold rates to defend the dinar: correct orthodoxy, strangled growth.

The loop's answer: owned export engines earn ~$117M/yr of gross exports outside the Fed cycle; the substitution legs take $152M/yr out of the dollar invoice directly; reserves rise, the rating runs B-/Caa1 → BB-, and the refinancing premium shrinks for the state with its own hard-currency engines. Not defiance of the Fed — independence from the channel.

The household ledger — what this does to the cost of living

A state program that does not make life cheaper for its people is decoration. Each line, computed from the same primitives as the rest of this document; the quarterly price index in Gafsa and Gabès is the published enforcement: **the price of protein and vegetables must bend downward within 24 months of the first harvest.**

HOUSEHOLD LINE	TODAY	AFTER THE LOOP
Water	$1.00-1.50/m³ (SONEDE produced)	$0.35/m³ coastal · $0.34/m³ inland
Protein	$15-25/kg imported	$5-10/kg local
Terfas	export €20-40/kg	one third stays home at ≤€12/kg, contractual
Health	the disease's bill	$2,592 per healthy year vs the $4,000/DALY standard
Energy (grid)	$0.110/kWh industrial	$0.031/kWh for the loop's loads; tariffs stay state policy
Work	28% unemployment (54% women graduates)	workshop, nursery, ponds, plants, crews

The energy ledger — what power costs, per party

Anchors cross-checked: Rw10 (0.352 TND/kWh industrial, 19 TWh), Pg14 (the government's own 1,100 GWh = 5% of production, ~$125M/yr gas saving), Rc13 ($31/MWh inside the $24-35 historical bid band), Pg15 (fuel imports $4.895B, WITS-verified).

PARTY	TODAY	AFTER THE LOOP	THE DELTA
The loop's industries	grid $0.110/kWh	**$0.031/kWh** on the loop's PPA	−72%
The state	19-22 TWh/yr, ~95% gas, $4.90B fuel bill	876 GWh/yr solar (4.0-4.6% of production)	−$106-125M/yr gas imports ($114-142/MWh band)
The household	subsidized STEG tariff	the cost stack beneath the tariff falls	the inflation channel throttled — every dinar protected

The frontier — where the loop reaches, and where it honestly stops

The complete household budget, swept — nothing hidden, nothing promised beyond the frontier.

HOUSEHOLD LINE	STATUS	THE HONEST NOTE
Electricity (household tariff)	**indirect**	household tariffs stay subsidized STEG policy; the loop lowers the cost stack they sit on and the subsidy burden the state carries

HOUSEHOLD LINE	STATUS	THE HONEST NOTE
Cooking gas (the bonbonne)	**out of scope**	the loop displaces grid gas, not bottled LPG; the bonbonne subsidy is a separate state decision
Transport fuel	**out of scope today**	the hydrogen leg serves ammonia for the fertilizer chain; transport fuels are a later policy question
Wheat and bread subsidy	**out of scope**	the protein stack complements the grain program; it does not replace it
Water	**covered**	cost floor $1.00-1.50 -> $0.35 coastal / $0.34 inland
Health	**covered**	$2,592 per healthy year vs the $4,000 standard
Protein	**covered**	$5-10/kg local vs $15-25/kg imported
Work	**covered**	the workshop, nursery, ponds, plants, crews where unemployment is 28% (54% for women graduates)
Housing	**out of scope**	the loop does not build housing; named so it is not promised

The water stack — line-by-line, with each credit's failure mode

The $0.347/m^3 audit: the base is measured anchors; each credit is a loop-internal yield with a named failure mode; the degraded cascade still clears the $0.68 gate.

LINE	$/M^3	PROVENANCE	WHAT MAKES IT FAIL
Base — solar energy 4 kWh/m^3 at the loop's own PPA + capex + O&M	$0.567	MEASURED anchors (Rc13/Rw2/Rw3)	holds — if even the PPA lapses, grid power takes the base to $0.88, still under SONEDE
DC reject heat → RO feed preheat — one seawater pipe cools the servers, then arrives warm at the membranes (SEC −2.5%/°C)	−$0.007	DERIVED — Leg 03 + Leg 01, same pipework	fails only if no compute anchor tenant — and costs $0.007
Kiln/exotherm preheat — the acid plant and board kilns warm the same feed	−$0.000	DERIVED — Leg 04/05 waste heat	negligible by design — carried at zero margin
Brine valorization — NaCl by solar-crystallization ponds (free sun); Mg(OH)&sb2; by lime precipitation from the loop's own kilns (134 t/yr lime, 0.14 GWh/yr — no evaporation), 100 kt to the board's fire retardant + 40 kt surplus sold at the park gate	−$0.320	DERIVED — process specified, CARMEn-anchored cost (Pg-anchor: 2025 techno-economic study)	if the Mg(OH)&sb2; of-take is not signed by FID, the credit degrades to −$0.03 (disposal avoided only) — the biggest single sensitivity
Shared civil works — one intake, one outfall, one grid connection serves RO + DC + metals in the same park	−$0.032	DERIVED — 12% capex sharing	if the park phases one module at a time, halve it to −$0.016
Dispatch value — the desal's own flexibility follows the loop's own solar curve	−$0.015	DERIVED — 20 GWh shifted	fails to zero if the STEG contract does not reward flexibility — contractual, not hoped

LINE	$/M³	PROVENANCE	WHAT MAKES IT FAIL
ESG financing delta — the health leg's public KPIs reprice the loop's own debt (100–150bp)	−$0.026	DERIVED — nations_v3	fails to zero if lenders do not accept the KPI track record — bankable-region fallback keeps the module alive
Brine pump-as-turbine — the desal's own residual pressure	−$0.002	DERIVED — 2.5 bar × 75%	fails to zero — immaterial
Integrated	**$0.164/m³**	**13% of SONEDE's $1.00-1.50 (Pg16)**	**the degraded cascade: if every soft credit fails, water still lands at $0.521/m³ — under SONEDE's own $1.00–1.50, and the $0.68 gate still clears with margin. The credits are the margin, not the plan.**

The Darwinian ledger — why business as usual loses

Protein built the human brain (Pg27); the same lever, applied on purpose, is the plan's deepest one. The business-as-usual ledger is the fear-facts: what “keep doing what we do” actually buys.

FACT	THE NUMBER	WHAT IT MEANS FOR THE NEXT GENERATION'S BRAINS
Fluorosis belt	24% dental fluorosis (Rw1)	high-F water ↔ lower child IQ (Pg28) — BAU taxes the cortex before it forms
Protein poverty	first 1,000 days (Pg29)	permanent cognitive deficit — the trees version, built child by child
Brain drain	tertiary emigration among world's highest (Pg30)	the country exports its cortex like its phosphate
The proof	InstaDeep sold for $680M	the cognitive gold was real — and sold, not kept
Debt	26.2% of export revenues to service (Pg13)	the budget cannot buy the interventions the brain needs
The idle cohort	28% unemployment; 54% women graduates (Rm19)	built brains, no environment — adaptation = exit
Inflation	5.15% on $4.90B fuel + $3.00B food	the first cut is diet quality — the Fed reaches the children's IQ

SELECTION PRESSURE	WITHOUT THE PLAN	WITH THE PLAN
Water scarcity	selects emigration — the skilled leave first	the mesh re-specifies the environment; adaptation becomes staying
Protein poverty	caps cognition permanently	ponds + canteens + quota reverse the pressure
The debt cycle	the surplus is eaten; the economy cannot invest	owned exports + substitution cut the invoice
Institutions	gates stay closed; skills emigrate through them	public gates select for competence, in public

The open-field plan — terfas, hectare by hectare

RUNG	HECTARES	SEEDLINGS	PRODUCTION	VALUE	OPENS ON
Gate (yrs 1-4)	100	600,000	≥200 kg/ha	first fruit ≥€20/kg	3-season record + contract
Scale (yrs 4-8)	2,500	15M	940 t/yr	€16544-29704M/yr	nursery output
Ceiling (yrs 8+)	3,000	18M	1,128 t/yr	inside 1,000-1,500 t/yr market cap (Rc8)	market offtake signed

6,000 mycorrhized seedlings/ha (Rc16) — the nursery is the first asset: 5M seedlings/yr of local-ecotype capacity, ~200 nursery + ~1,000 field + ~783 seasonal jobs at scale. Local wild genetics, not the imported soil biology that failed in Saudi Arabia (Rc17).

The AI spine — the system's nervous system, and its boundary

The AI runs the measurement, the routing and the publication — never the decisions.

COMPONENT	WHAT IT DOES	WHAT IT IS WORTH
The dispatcher	routes every joule and m³ in real time: solar → desal → electrolyzer → ponds → DC cooling, following the solar curve minute by minute	worth the 20 GWh/yr of dispatch value — the AI is what collects it
The scoreboard	computes and publishes every gate number automatically — filter uptime, urinary F, DSCR, yield records — tamper-evident, no human can edit a gate	the public contract, machine-audited
The price index	tracks the Gafsa/Gabès household basket quarterly and publishes the bend — the people's enforcement metric, computed not curated	the number that proves the prices bend, or proves they do not
The cohort engine	runs the 3,000-family health baseline, dose-outcome analytics and anomaly flags for the clinical team	the DALY ledger becomes a live dataset
The dispatcher twin	a digital twin of the loop (energy, water, brine, heat) that simulates before any physical change and audits after	the falsification ladder, automated
The boundary	the AI never sets a gate, never signs a contract, never prices a tariff — it measures, routes and publishes. The decisions stay human, in public.	the guardrail that keeps the nervous system a servant, not a master

The results framework — what a World Bank board reads first

The scoreboard IS the M&E system (PforR class, Pg23).

INDICATOR	BASELINE	TARGET	DATA SOURCE
Filter uptime, pilot households	0% (filters not installed)	≥70%	scoreboard, live
Urinary fluoride, pilot children	baseline to be measured (Day 1-30)	−20%	cohort engine

INDICATOR	BASELINE	TARGET	DATA SOURCE
Household price index, Gafsa/Gabès	baseline Q0	bend downward ≤24 months after first harvest	price index, quarterly
Water produced cost	SONEDE \$1.00-1.50/m^3 (Pg16)	≤\$0.35/m^3 integrated	plant telemetry
Energy recovered inside the loop	0 GWh/yr	33.9 GWh/yr + 20 GWh dispatch	dispatcher ledger
Gas imports displaced	\$4.895B fuel bill (Pg15)	−\$106-125M/yr	WITS Comtrade, annual
Uranium recovered from the acid stream	0 (lost to the sea)	≥85% recovery gate	assay chain
Terfas hectares planted	0 ha	100 ha gate → 2,500-3,000 ha	nursery registry
Jobs in the basin	28% unemployment (54% women graduates)	workshop → nursery → plants → crews	employment registry

The people's energy is electrons, not hydrogen — and the H&sb2; lives inside the loop

The loop's H&sb2; (2.63 kt/yr) is an industrial molecule: per kWh of heat, green H&sb2; at \$0.21 is 7× the loop's \$0.031/kWh electron; blending caps at 5-20% by volume (Pg20/Pg21/Pg22); the whole H&sb2; volume equals 1.8% of grid-gas demand. The molecule goes where only a molecule works: ammonia for the GCT chain, inside the loop. the electrolyzer sits on the desal line, and everything it touches stays in the loop: the water it splits is the SWRO permeate (0.04% of the 150,000 m^3/d output); its un-split reject stream is clean water that returns to the product line; its oxygen by-product aerates the spirulina ponds and greenhouses; its stack heat (15–20% of input) joins the kiln and DC reject heat on the desal preheat circuit; and the minerals concentrated in its flush stream join the brine valorization line. The electrolyzer is not a consumer of the loop — it is another recovery point inside it.

The total routing — every stream gets a home

The ponds do *not* drink brine: SWRO brine is 55-70 g/L, past Arthrospira's ~30 g/L ceiling (Pg31); they drink the desal permeate and breathe the electrolyzer's oxygen. The full inventory:

STREAM	PRODUCED BY	ITS HOME	THE NUMBER
Solar electrons	500 MW fleet	desal, metals, DCs, electrolyzer, pumps — routed by the dispatcher	876 GWh/yr
SWRO permeate	coastal desal	city offtake · electrolyzer feed · ponds · greenhouses	150,000 m^3/d
SWRO brine	coastal desal	salt + magnesium valorization, evaporated on the loop's own free heat	122,727 m^3/d → 58 kt Mg/yr
Brine bittern	valorization	Mg(OH)&sb2; fire retardant → **the board's own chemistry**	140 kt/yr = 140% of the board's need

STREAM	PRODUCED BY	ITS HOME	THE NUMBER
Electrolyzer oxygen	H&sb2; plant	spirulina pond + greenhouse aeration	21 kt/yr
Electrolyzer hydrogen	H&sb2; plant	ammonia → the GCT fertilizer chain	2.63 kt/yr
Electrolyzer reject water / heat / flush	H&sb2; plant	product line / desal preheat / brine line	nothing leaves
DC reject heat	compute park	the same seawater pipe, arriving warm at the membranes	$-\$0.007/\mathrm{m}^3$
Kiln + acid exotherm	board kilns + metals plant	desal preheat + greenhouses	30 GWh/yr
CO&sb2; (kiln + power)	combustion streams	mineralized into board (2 Mt PG) · pond carbon	540 t CO&sb2;/yr to the ponds
Phosphogypsum	GCT stack	fire-rated board — with its retardant from our own brine	2 Mt/yr
Acid stream	GCT wet-process acid	U + heavy REE extraction; raffinate back to fertilizer	716.5k lbs U&sb3;O&sb8;/yr
Municipal return flows	city mesh	30%+ reuse · pump-as-turbines on the descent	3.9 GWh/yr
Sludge / digestate	wastewater + organics	biogas for kilns and greenhouses	booked option, honestly labeled
Slaughterhouse bones	local slaughterhouses	bone char → the fluoride filters; **spent char → phosphate soil amendment for the fields**	filter loop closes into fertilizer
Spent pond medium	spirulina harvest	nutrient-rich → greenhouse fertigation	nothing discarded
Ammonia fertilizer	GCT chain	the terfas fields' host plants + greenhouse soils	the truffle grows on the loop's own nitrogen
Compute services	colo park	export — the data product	tenant-gated
Ra-226 / Po-210	acid stream radiology	stays in the gypsum/board matrix — the radiology gate	below the 1,000 Bq/kg exemption
Board offcuts	board line	recycled into new board	zero waste
Sodium chloride	brine line	industrial offtake at the park gate	nothing hauled
Halophytes / Artemia on brine	brine line	future option, honestly labeled — not a current credit	the only stream awaiting its consumer

the ponds drink the desal permeate — ~{:.0f} m^3/d, {:.2f}% of the 150,000 m^3/d output — and breathe the electrolyzer's oxygen; the brine (55–70 g/L, far past the ~30 g/L ceiling the organism tolerates) is routed to salt and magnesium valorization instead. Halophytes and Artemia on brine are booked as a future option, honestly labeled — not a current credit.

The fuel invoice, honestly split — the 500 MW is the grid's medicine, not the whole prescription

of the $4.895B fuel invoice, the 500 MW fleet displaces only the power-sector gas slice — ~$0.75B/yr (DERIVED from the energy balance) — by $106-125M/yr, which is 14.1-16.7% of that slice. The rest of the invoice has its own answers, not solar's: the bottles (~$0.55B of LPG, the bonbonne) leave the market as the people shift to electrons — induction on the loop's own surplus — and the biogas pilot; transport fuels (~$2.6B) are the honest frontier. The 500 MW is the grid's medicine, not the country's entire prescription.

The mountains, honestly analyzed

Fog collection is real where fog lives (10-20 L/m²/day, Pg33) — Tunisia's fog lives in the already-humid NW; the truffle mountains are the arid interior, and cloud seeding has no proven routine gain (Pg32). What the mountains give the loop: the mountains enter the loop as storage, rainfall and free cooling — and as the land where the truffle belongs (freeing the lowlands that grow today's food). Fog harvesting from pipe meshes stays off the program: the moisture is not there to collect, and the honest scientist does not plant a net under a cloudless sky.

The AI regulation — five layers, down to the drippers

The dispatcher is a model-predictive controller on a 15-minute receding horizon over the mass balances (Lexicographic: safety > contracts > economics); the twin trains it offline and audits it online, with a Kalman filter for unmeasured states and a published prediction-error KPI; node controllers stay PID-class with hard-wired interlocks; the governance layer never delegates a gate.

LAYER	WHAT IT DOES	THE MATHEMATICIAN'S NOTE
Layer 0 — the instrumented field	every node is a sensor + an actuator: membrane pressure and conductivity, brine concentration, pond pH and temperature, electrolyzer stack voltage, kiln thermocouples, PAT speed, reservoir levels, household-filter telemetry, the grid tie, the weather station	the stream inventory above is the measured state — the mass balances are the constraints the controller must never violate
Layer 1 — node controllers (edge, millisecond-to-second)	local closed loops of PID class: membrane pressure, pond dosing, kiln temperature, pump speed — with hard-wired safety interlocks	functional safety is never on the AI: a controller that fails safe is a relay, not a model
Layer 2 — the dispatcher (model predictive control, 15-minute receding horizon)	a multi-variable MPC: state = solar forecast, reservoir inventory, brine concentration, electrolyzer turndown, DC load, grid price, offtake commitments; decisions = pump flows, setpoints, brine routing, dispatch; forecasts 36–48 h weather, 24 h load	objective: minimize loop energy cost + degradation + gate-violation penalties, subject to mass balances, storage states and contract floors — lexicographic, never weighted: safety > contracts > economics. Its measured output is the 20 GWh/yr dispatch value and the routing-dependent water credits
Layer 3 — the digital twin (simulation, learning, audit)	a physics-calibrated model of every balance: trains the dispatcher offline, audits every decision after the fact, simulates before any physical change	unmeasured states (membrane fouling, pond bloom state) are estimated with a Kalman-class filter from pressure-flow and optical residuals; the twin’s

LAYER	WHAT IT DOES	THE MATHEMATICIAN'S NOTE
		prediction error is a published KPI — the AI’s accuracy is measured, not claimed
Layer 4 — the governance layer (the boundary, unchanged)	the scoreboard computes and publishes every gate number from Layer 0 telemetry with provenance; the price index; the results framework; anomaly flags to humans	the AI never sets a gate, never signs, never prices — the falsification ladder is mechanical thresholds, not a learned model. The nervous system is a servant, not a master

NODE	SENSES	CONTROLS	TIMESCALE	LAYER
PV field	irradiance, string current, panel temperature, soiling sensors	inverter setpoints, curtailment, cleaning scheduling	seconds–minutes	L1–L2
Desal trains	feed pressure, permeate conductivity, ERD pressure, CIP condition	high-pressure pump, ERD bypass, antiscalant dosing, CIP cycles	seconds–hours	L1–L2
Membrane health	pressure-flow residuals (fouling inferred by Kalman filter)	maintenance scheduling, train rotation	days	L3
Brine line	concentration, temperature, Mg saturation	evaporator feed rate, Mg(OH)&sb2; precipitation setpoints	minutes	L1–L2
Electrolyzer	stack voltage, current, purity, water feed	turndown, ramp rate, O&sb2; vent routing to ponds, reject routing	seconds	L1–L2
Board kilns	temperature profile, feed rate, offcut fraction	burner setpoints, retardant dosing, offcut recycle	minutes	L1–L2
Metals plant	SX stream assays, raffinate quality	extraction/stripping setpoints, raffinate return to fertilizer	minutes–hours	L1–L2
DC park	cooling-loop temperature, PUE, workload mix	seawater flow to racks, batch-workload scheduling to solar peaks	minutes	L2
Water mesh	reservoir levels, pump status, pressure heads, PAT speed	pump scheduling, zone valves, PAT bypass, the solar-following curve	15-minute dispatch	L2
Spirulina ponds	pH, salinity, temperature, optical density	aeration timing, CO&sb2; injection, medium exchange, harvest scheduling	hours	L1–L2
Greenhouses	EC, pH, humidity, CO&sb2;, soil moisture	fertigation dosing (spent medium), ventilation, CO&sb2; enrichment from the kiln stream	minutes	L1–L2
Establishment drip — to the emitter	per-zone capacitance soil-moisture probes, per-zone flow meters, weather forecast	zone valve actuators per management block, pump pressure, fertigation dosing — emitters stay passive pressure-compensated; clogged drippers detected from per-zone flow residuals	hourly–daily	L1–L2
Household filters	usage telemetry, replacement flags	fleet logistics: replacement routing, spare inventory, the pilot's uptime KPI	days	L2–L4

NODE	SENSES	CONTROLS	TIMESCALE	LAYER
Grid tie	import/export price, solar forecast, offtake commitments	import vs. dispatch decisions	15-minute	L2
Scoreboard + cohort	every KPI stream above, provenance-hashed	gate computation, price index, anomaly flags to humans	continuous	L4

what the AI is honestly worth: the 20 GWh/yr dispatch (~$0.8M/yr) + the routing-dependent water credits it executes (~$0.05-0.08/m³ — DC heat, kiln heat, brine heat, dispatch) + a DERIVED 1–3% park OPEX from multi-variable coupling no manual schedule can follow — and the scoreboard, which is the program's entire credibility. if the AI is down, the loop does not stop: Layer 1 runs every node in safe, suboptimal mode — the degraded cascade already computed ($0.521/m³ water, gate still clear). AI uptime is a KPI, not a dependency.

The irrigation — to the dripper, honestly

the fields drink at establishment, then live on rain — and the greenhouse loop runs on the pond's own spent nutrients. The irrigation system is designed the way the truffle itself is: drought-proof by architecture, not by subsidy. Establishment: 624 m³/ha/yr for the first two years → 4,274 m³/d for 2,500 ha, from inland brackish water and reuse; then rainfed.

Cloud generation — the honest version

our 12 km² array absorbs ~156 MW more sunlight than the bare soil (albedo Δ0.05) — a persistent local heat source, plus the ponds' evaporation placed upwind of the fields. Li et al. (Pg34) measured the class of effect at continental scale; at ours it is a designed microclimate, not a rain machine: soil shaded by panels retains moisture, the updraft adds marginal convective activity, and the dew that follows is free water on the host plants. The honest claim is directional and modest — and it costs nothing extra: the array was being built anyway.

The mountain portfolio — if mountain agriculture is needed, the version that scales

CROP / SYSTEM	HECTARES	WATER	VALUE	THE SCALE LOGIC
Terfas belt (arid jebels)	2,500-3,000 ha	rainfed + micro-catchments + dew	€15-45M/yr at scale (market-capped)	already the design: the drought-proof luxury crop on land nothing else farms
Cactus pear (opuntia) belt	consolidate + expand the existing ~600k ha national belt	zero irrigation; 300-500 mm survives on 150	fruit at 20-40 t/ha for the basin's price index + cladodes as livestock fodder	the arid mountain staple — Tunisia is already a world producer; the program adds value chains, not water
Medicinal & aromatic plants (rosemary, thyme, oregano)	5,000-10,000 ha across the Dorsale	rainfed; 300-500 kg/ha dry	~€8-20M/yr export at €4/kg dry	the highest value-per-kilo crop that needs no water at all — the mountain cash crop #2

CROP / SYSTEM	HECTARES	WATER	VALUE	THE SCALE LOGIC
Agrivoltaic fodder (sulla/alfalfa under the PV field)	up to 1,250 ha of shaded soil under the 500 MW array	the panels' shade halves evaporation; no added water beyond the field's own rain	fodder for ~5,000-10,000 head; the flock fertilizes the field	the loop's livestock leg: sheep graze solar parks already (Spain/US precedent) — the AI routes the grazing schedule
Rainwater harvesting (swales + terracing)	2-3% of mountain catchment area	each ha of swales captures ~3,000 m^3/yr at 300 mm rainfall	$200-500/ha — an order of magnitude under drip-system capex	the mountains' real water system: catch the rain that already falls there, at scale, without pipes

mountain agriculture at scale means putting each crop where its water cost is zero: terfas, cactus, MAPs and agrivoltaic fodder on the mountains' own rain and the loop's own shade — while wheat, vegetables and orchards stay on the lowlands and in the greenhouses where water is abundant. The mountains feed the people what only mountains can grow; the pipes carry water only where gravity and economics both say yes.

The growroom at landscape scale — climate steering toward bio-recovery

Measures: Atmospheric transect: RH / temperature / dew-point stations up the slope at the mist corridor, leaf-wetness sensors, 2-3 sonic anemometers for the anabatic flow; Soil grid: capacitance moisture probes per management zone (the drip grid doubled as the climate grid), soil temperature; The sky: satellite NDVI every 5 days into the twin, rain gauges along the corridor, pond evaporation pans; The park: panel albedo/soiling sensors (the field's reflectivity IS a climate actuator), pond water temperature

Steers: The mist corridor: fog nozzles, anabatic hours only, virtual-temperature-positive dosing — 2% of desal flow; The shade: the PV array's own footprint: shaded soil retains moisture; cleaning schedule shifts albedo; The pond surface: evaporation timing follows the wind — the ponds breathe upwind of the fields on purpose; The swales: rainfall harvested at contour, released by zone valves into the host-plant belt

Objective: the bio-recovery term in the dispatcher’s objective: maximize dew days + soil-moisture persistence above wilting point + NDVI slope — subject to the energy budget, the thermals remaining buoyant, and the rain-gauge referee. The twin adds a soil-water-balance module (Penman-Monteith evapotranspiration, standard equations, computed in code) so every steering decision has a predicted and a measured outcome.

The spiral: the feedback the steering rides: mist → dawn dew → soil moisture → host plants establish → transpiration raises local humidity → more dew — the humidity loop that turns a slope back green, one feedback cycle at a time. The KPIs published: dew days per month, days above wilting point, NDVI slope — bio-recovery, measured like a growroom, audited like everything else.

The invisible drop — and the effects ledger, direct and side

the drop → microbial pulse → nutrient mineralization → plant-available N and P → host-plant growth → nectar and pollen → the insects: wild bees at 1,000-2,000 flower visits a day, ants engineering the soil, beetles on the wind → the host plants set seed (Helianthemum is insect-pollinated) → the mycorrhizal trade persists → the truffle → the harvest → the price index. Every dew morning the mist buys is 10&sup8; cells per square centimeter of leaf waking up — and the corridor's vegetation is the pollinator highway the truffle's own host plants depend on. One invisible drop feeds the microbes that feed the plants that feed the insects that pollinate the plants that feed the truffle that feeds the people.

and the drop re-enlists the army Tunisia lost: the right-of-way vegetation is the refuge for the beneficials — ladybirds (5,000 aphids each in a lifetime), lacewings, hoverfly larvae (400-600 aphids each), and the parasitoid wasps Tunisia already proved at national scale in the date oases (Trichogramma against the date moth). Every predator the corridor hosts is a pesticide that was not bought, not sprayed, and not inhaled by a child — biocontrol is the health ledger's quietest leg.

EFFECT	CLASS	THE MODEL	THE INSTRUMENT
Mist vapor transport	DIRECT	Gaussian plume + anabatic advection	sonic anemometers, RH transect
Dew deposition on host plants	DIRECT	surface energy-balance dew model (T&sb2;surf < dew point)	leaf-wetness sensors, dew gauges
Soil moisture gain, corridor zone	DIRECT	Penman-Monteith water balance	capacitance probe grid
Seedling establishment survival	DIRECT	water-stress survival curves	nursery registry, field counts
PV-shade moisture retention	DIRECT	shade fraction × PET reduction	soil probes under panels
Thermal buoyancy (the constraint)	DIRECT	virtual-temperature profile	T/RH + wind stations on the slope
Microbial Birch pulse	SIDE	rewetting respiration curves	soil CO&sb2; efflux chambers
Beneficial insect recruitment	SIDE	habitat-area × predator-prey dynamics	sticky traps, acoustic/AI counters
Pollinator visitation → host seed set	SIDE	pollination-saturation curves	visit counts, seed counts
Precipitation return	SIDE — ZERO-credited	cloud microphysics (the least-solved field, Pg32)	rain-gauge network
Pesticide displacement → child exposure	SIDE	exposure-response from the cohort	urinary pesticide metabolites
Atmospheric dust & PM scavenging	SIDE	wet-deposition scavenging coefficients (dew/fog wash)	PM sensors along the corridor
Air quality, basin-wide	DIRECT	emissions inventory → dispersion model	PM2.5/NOx/SO2 grid + the cohort's respiratory KPIs

The land, projected — the Darwinian view of the environment itself

LAYER OF LIFE	WITHOUT THE PLAN	WITH THE PLAN
Soil biology	the seed bank dies with the soil — desertification's positive feedback: less vegetation, less humidity, less dew, less vegetation	the Birch pulse wakes with every dew morning; the microbial city rebuilds; 10 km of hyphae per gram of soil come back to trade

LAYER OF LIFE	WITHOUT THE PLAN	WITH THE PLAN
The insects	beneficials stay locally extinct; the pesticide treadmill accelerates; pollinators thin, host plants set less seed	the right-of-way is the refuge: ladybirds, lacewings, hoverflies, Trichogramma's cousins re-recruit; every predator is a pesticide not bought
The dew economy	bare, crusted ground sheds what little rain falls; the drop dies on arrival	the mist corridor turns the dawn into a harvest: dew on host plants, soil moisture banked, the spiral compounding
The truffle belt	the wild harvest shrinks; the mycorrhizal network starves	the belt establishes on the corridor's moisture: 100 ha, then 2,500-3,000 ha, on the land the humidity itself prepared
The frontier	desertification marches downhill — the slope's fitness landscape selects for stone	each humidified season pushes the establishment frontier a little further upslope — the landscape's selection flips from bare ground to vegetation

the Darwinian projection of the whole machine: the same land, two futures — one where the environment keeps selecting for stone, and one where a 2% mist margin re-engineers the fitness landscape so that life out-competes bare ground. The honest bound stands: the corridor transforms a belt, not a mountain range — the wider slopes change only through the slow, compounding spiral and the rain cycle. We plant the drop; the land grows the rest.

The nature ledger — every class of credit, not just carbon

SPECIES / ASSET	THE HONEST NOTE
Wild bee community	~700+ species documented in Tunisia; the corridor's nectar belt is their refuge and the truffle host plants' pollination engine
Egyptian vulture (globally endangered)	breeds in Tunisia; scavenger of the open steppes the corridor stabilizes
Dorcas gazelle	endangered; the restored steppe of the south is its historic range
Cork oak + orchid flora (NW)	the humid mountains' own endangered flora — the value-chain program that ends over-harvesting
Medicinal & aromatic plants	wild-rosemary pressure relieved by the cultivated MAP belt — the market saves the wild stock
Addax	honestly named: locally extinct in the wild — NOT claimed; restoration is a later, separate question

the nature ledger: the corridor's ~50 km² of restored belt at maturity earns **$0.0-0.2M/yr** of biodiversity credits at the nascent-market band ($5-50/ha, DERIVED — booked at the low end; TNFD disclosure framework, Pg37; GBF Target 19, Pg36) and unlocks a **debt-for-nature swap**: $500M of the state's debt refinanced against measured nature outcomes — ~$7.5M/yr of coupon savings, the Gabon/Ecuador instrument (Pg38). The MRV system already exists: the acoustic counters, the NDVI feed, the soil grid — the same instruments that steer the mist also certify the credits. Not counted in the FX bridge (nature credits are not FX-grade exports): this is the program's second ledger, and the scoreboard publishes it the same way.

The credit spectrum — every financial credit the country earns

CLASS	WHAT IT IS	THE HONEST NUMBER	STATUS
Carbon credits	280 kt CO&sb2;/yr avoided + mineralized + sequestered (solar displacement, the board, the restored belt's soil)	$1.4-4.2M/yr at $5-15/t — the compliance/voluntary band; VERIFIABLE, Phase 0 baseline opens it	creditable
Biodiversity credits	the restored corridor belt, certified by the same sensors that steer the mist	$0.03-0.25M/yr at the nascent low band	creditable, nascent
Debt-for-nature swap	$500M of sovereign debt refinanced against measured nature outcomes	~$7.5M/yr coupon savings (Gabon/Ecuador precedent, Pg38)	instrument, negotiable
ESG-linked debt discount	the health KPIs reprice the program's own debt	100-150bp on the program portfolio — already counted in the water stack	booked
Concessional windows	GCF, GEF, Adaptation Fund, EU Global Gateway co-financing — the program's public KPIs are the eligibility	blended-finance tranches on solar/desal/health — applied, not assumed	grant/soft-loan
Rating-linked spread compression	B- to BB- sovereign re-rating on the measured path	the state's upside of $1.2-2.0B/yr on the refinancing share — NOT booked in the program ledger	the state's harvest

the credit spectrum is the program's third ledger — carbon, biodiversity, the swap, concessional windows — each class bookable, none double-counted against the FX bridge, each with its MRV already instrumented. The country's nature, measured by the country's own sensors, becomes the country's credit line.

The credit rating, projected — the way the agencies rate

FACTOR	TODAY	AT FULL PHASE
Reserves / import cover	4.5 months today (measured)	FX accumulation at full phase: ~$1.0-1.4B over the 5-year build → 6+ months — the S&P reserves-adequacy factor moves
External position	CA deficit ~-3.3% of GDP	the loop narrows it by ~0.5 pts (279M/yr) → ~-2.5-2.7% — the agencies rate the PATH, not just the level
Growth	~1.9% today	+0.5 pts from the loop + 1.2 pts from the peace dividend → 3-4% — the strongest single upgrade factor for B-rated sovereigns (Pg40)
Fiscal	subsidy burden + debt service 26.2% of exports	the substitution legs + the swap's coupon savings + the ESG delta relieve the budget on the same lines
Institutional predictability	policy volatility is the chronic negative	the public scoreboard, the gates, the apolitical program — a measurable governance signal the methodology rewards

The path: B- → **BB- / Ba3 within 3-5 years** (reserves 6+ months, CA ≤-2.5%, growth ≥3% sustained — all inside the measured numbers) and **BB / Ba2 in 5-7 years** with the swap and the nature ledger. Investment grade is NOT claimed: that requires debt/GDP ≤60% and a decade of institutional track record — the honest frontier.

the spread arithmetic, stated for what it is: B- to BB- sovereign compression is ~300-500bp; on the refinancing share of the $40.5B stock that is the state's upside of $1.2-2.0B/yr — NOT booked in the program's ledger: only the swap's measured $7.5M/yr and the program-debt ESG delta are credited. The rating is the state's harvest; the program only plants.

Air quality — the air the children breathe

The board plant caps the phosphogypsum stack dust; the solar leg displaces the grid's gas burn; the biocontrol corridor replaces pesticide drift; and the dew the mist buys scavenges PM from the air (wet deposition). Measured by the PM2.5/NOx/SO2 sensor grid, the dispersion model and the cohort's respiratory KPIs — air quality enters the scoreboard the same way the water does.

The last inventory — nothing forgotten, nothing hidden

ITEM	THE GAP IT CLOSES	ITS HOME	STATUS
Pond blowdown (the salt bleed)	the spirulina ponds' own concentrate, when the recycled medium's salts saturate	routed to the brine line — the pond's salt joins the valorization stream; the water loop closes, the salt loop closes	routing row, added
Construction carbon & energy payback	the embodied footprint of building the park (PV, concrete, steel) vs the operating recovery	PV energy payback ~1-2 years (industry standard, DERIVED); the park's embodied carbon is repaid by the displacement + mineralization inside the first operating years — computed in the Phase 0 LCA item	LCA item in Phase 0
The risk-transfer layer	drought years, PV underperformance, price shocks — the hazards the gates alone cannot absorb	parametric insurance on the mist corridor's target (rainfall index) and the solar PPA (irradiation index); the degraded cascade already computed is the self-insurance floor	insurance, bookable
Nuclear export compliance	U&sb3;O&sb8; leaves the country — the chain of custody the IAEA/NPT regime demands	the CNSTN gate already named extends to export controls: buyer compliance, safeguards accounting, transport certification — the Cameco-class compliance chain, cited as a gate, not an afterthought	compliance chain, gate
The women's leg	54% of women graduates unemployed — who exactly gets the jobs must be designed, not hoped	the nursery, the workshop, the pond crews and the health workers as women-led cooperatives in the blockade towns — the quota principle already proven in the terfas domestic clause, applied to employment	cooperative structure, contractual
The AI's own audit	the dispatcher's code and the twin's models must be auditable or the scoreboard's trust is borrowed	open models, versioned, the prediction-error KPI published, independent code audit before each phase — the nervous system is inspected like everything else it measures	code audit, gate

The fauna water ladder — the earwig's drop, as the unit of the whole scale

SPECIES	MASS	WATER / DAY	THE NOTE
The earwig	0.1 g	~0.5 µL/day (measured class)	one 50 µL drop = 100 earwig-days — the owner's own observation, as the unit of the whole scale
The wild bee	0.1 g	~30 µL/day (measured)	a 10 mL/m² dew morning serves ~300 bees per square meter
The desert lizard	20 g	~1.4 mL/day (FMR)	dew-licking Acanthodactylus — the Saharan micro-fauna's classic water behavior
The sandgrouse	250 g	~10 mL/day (FMR)	the Sahara's dew-drinker; chicks absorb water through belly feathers — the corridor's fog mornings are its nursery
The fennec fox	1.2 kg	~36 mL/day (FMR)	the desert's smallest canid — water from prey, dew and the corridor's troughs
The Egyptian vulture	2 kg	~54 mL/day (FMR)	globally endangered, breeds in Tunisia — the corridor's carrion-cycling scavenger
The Dorcas gazelle	17 kg	~298 mL/day (FMR)	endangered; its historic southern steppe is the restoration belt itself
The Barbary sheep	50 kg	~706 mL/day (FMR)	the mountains' own ungulate — the corridor's upslope fringe is its range
The loop's flock	60 kg	~800 mL/day	the agrivoltaic sheep: the mist's 2% is its drinking trough, and its grazing is the land's firebreak

one average day's dew on the corridor — 51 m^3, the same water as one house's annual use — rations across the whole ladder: 10^{11} insect-days or 1.7 billion bee-days or a million vulture-days or 170,000 gazelle-days. The species that drink it are the ones Tunisia is losing: the scale that ends at the Dorcas gazelle begins at the earwig's half-microliter — and the mist corridor is the trough for every rung. Allometric spine: daily water = 0.123 x mass^0.8 mL (Nagy, Pg43), cross-checked against measured insect rates.

The spirulina health claim, at the evidence: and the health claim, downgraded to the evidence: the 2025 meta-analysis shows positive weight/MUAC effects in malnourished children (Pg41); the 2026 meta-analysis finds NO significant growth effect (Pg42). The page claims the FOOD, not the cognition — the cohort measures the rest, and the gates decide.

What this document is: a concept-level engineering design whose numbers are computed in code from sourced primitives — recomputed, corrected, and applied into the design. Free and public.

What it is not: a commissioned state program, a bank-grade feasibility study, an investment offer, or a claim that any of this has been built. Nothing here is advice to any government or any investor. Nothing replaces geological, radiological, geotechnical or EPC-level surveys. Verify every figure and citation before relying on any of it.

Works cited

1. [Rm1] Rare Earth Elements in Phosphate Ores and Industrial By-Products (Gafsa avg 322 mg/kg, peaks >1,700) — https://www.mdpi.com/2075-163X/15/12/1232...
2. [Rm2] Global phosphogypsum: generation, stockpiles and 60-85% REE partition to PG — https://www.mdpi.com/2071-1050/17/19/8828...
3. [Rm3] Gabès phosphate plants REE characterization: enrichments to 1,075 mg/kg; Ce 80-309 mg/kg; negligible Dy/Tb — https://www.researchgate.net/publication/352414505_Rare_earth_elements_characterization_associated_to_the_phos...

4. [Rm4] Characterization and Leaching Kinetics of REE from Phosphogypsum in HCl (1.65 mol/L; 65.6% ΣREE; 73.8% Y) — https://www.researchgate.net/publication/361001660_Characterization_and_Leaching_Kinetics_of_Rare_Earth_Elemen...
5. [Rm5] Recovering REE from phosphogypsum as a secondary resource: scale-up failure modes (filtration, silica gel, co-extraction) — https://www.chinanhd.com/recovering-rare-earth-elements-from-phosphogypsum-as-a-secondary-resource-pathway/...
6. [Rm6] PhosAgro pilot-scale REE recovery from apatite-derived PG (5 m^3/h — igneous, richer feed) — https://cris.vtt.fi/en/publications/pilot-scale-recovery-of-rare-earths-and-scandium-from-phosphogyps/...
7. [Rm7] Rainbow Rare Earths Phalaborwa: 2.2 Mtpa PG, 29.1% NdPr, pilot proven, OPEX <$13/kg (exception, not the rule) — https://www.rainbowrareearths.com/wp-content/uploads/2022/03/21-08-04-Sprott-Initiation.pdf...
8. [Rm8] 2024-2026 oxide prices: NdPr $64-103/kg; Dy ~$340/kg; Tb ~$1,800-1,875/kg; La $3.70/kg; Y <$5-10/kg; Ce ~zero — https://en.institut-seltene-erden.de/aktuelle-preise-von-seltenen-erden/...
9. [Rm9] Mixed rare-earth concentrate discount vs separated oxides (30-50%) — https://pubs.acs.org/doi/pdf/10.1021/acssusresmgt.5c00353...
10. [Rm10] Aclara: solvent-extraction separation plant Class-5 CAPEX $244M, OPEX $12/kg REO — https://investingnews.com/aclara-announces-update-on-its-rare-earths-separation-project/...
11. [Rm11] Tunisian PG radiological characterization: Ra-226 214-970 Bq/kg; U-238 88.5; Pb-210 755.4; co-precipitation risk — https://www.researchgate.net/publication/24188010_Characterization_of_phosphogypsum_wastes_associated_with_pho...
12. [Rm12] IAEA / EU BSS: 1 Bq/g (1,000 Bq/kg) exemption threshold for U-238/Ra-226; TENORM classification — https://www.mdpi.com/2071-1050/17/8/3473...
13. [Rm13] NORM disposal options and costs ($15-420 per 55-gallon drum) — https://www.ogj.com/home/article/17230971/norm-disposal-options-costs-vary...
14. [Rm14] China's 85-90% share of global REE separation/refining capacity — https://www.adlittle.com/sites/default/files/reports/ADL%20The%20rare%20earth%20imperative%202025.pdf...
15. [Rm15] Breakeven economics for non-Chinese REE projects (~$77/kg NdPr-equivalent; PG-grade requires >$150/kg) — https://www.elibrary.imf.org/view/journals/001/2026/072/article-A001-en.xml...
16. [Rm16] Hydrochloric acid prices: EU $124-150/t (Q4 2025); NA ~$350/t; China ~$15/t — https://chemtradeasia.sg/en/market-insights/hydrochloric-acid-supply-report-2024-2025-buyer-lessons...
17. [Rm17] GCT throughput (~6.5 Mt phosphate rock/yr; >10 Mt/yr PG at peak; 'hundreds of millions of tons' discharged at Gabès) — http://www.gct.com.tn/fileadmin/user_upload/Downloads/Rapport_Annuels/GCT-ANNUAL-REPORT-2012-PART1.pdf...
18. [Rw1] Largest SWRO plants: Ras Al-Khair 1,036,000 m^3/d; Taweelah 909,200 ($847M-1.2B); Sorek II 672,000; Rabigh 3 600,000 — https://ide-tech.com/en/project/sorek-b-desalination-plant/...
19. [Rw2] SWRO CAPEX $800-1,200 per m^3/day for mega-plants (>100k m^3/d) — https://supra-international.com/insights/seawater-reverse-osmosis-swro-desalination-technology-comprehensive-a...
20. [Rw3] LCOW at plant gate: $0.40-0.60/$m^3$ (Jubail 3A $0.41; Sorek II bid $0.32-0.45); energy share 40-60% of OPEX — https://pmc.ncbi.nlm.nih.gov/articles/PMC8953854/...
21. [Rw4] Pumping physics: 400 m lift = 1.09 kWh/m^3 at 100% eff; 1.45 at 75%; friction 0.5-1.5 kWh/m^3 over 300 km; realistic 2.0-2.5 kWh/m^3 — https://www.engineeringtoolbox.com/water-pumping-costs-d_1527.html...
22. [Rw5] Jubail-Buraydah IWTP: 587 km, 650,000 m^3/d, $2.26B ($3.85M/km); twin-pipe scale $5-7M/km — https://www.meed.com/saudi-arabia-awards-22bn-water-transmission-contract...
23. [Rw6] Delivered inland desalinated cost $1.10-1.50/$m^3$ (plant gate + transport + pumping energy) — https://investriyadh.ai/intelligence/water-scarcity-investment/...
24. [Rw7] Tunisian water tariffs: SONEDE 0.200-2.310 TND/m^3; irrigation 0.060-0.160 TND/m^3; groundwater 0.112-0.162 TND/m^3 — https://www.sonede.com.tn/tout-savoir-sur-leau/la-facture-deau-methodes-de-paiement/...
25. [Rw8] International transfer precedents: GMMR $33B (2.5-6.5 Mm^3/d, $0.28-0.62/$m^3$, fossil water); Riyadh levelized transport $0.42/$m^3$; <$0.20/m^3 is the agronomic ceiling — https://www.mei.edu/publication/whats-next-libyas-great-man-made-river-project/...
26. [Rw9] 24/7 firming: 3.03 GW @20% CF = 0.606 GW avg (zero margin); 9.15 GWh/night; BESS round-trip 85% → 12-15 GWh; $110-125/kWh → ~$1.5B; solar+storage LCOE $100-130/MWh — https://www.energy-storage.news/battery-storage-system-prices-continue-to-fall-sharply-bnef-and-ember-reports-...
27. [Rw10] Tunisia grid: STEG 5.9 GW, ~19 TWh/yr, 95% gas-fired, renewables <5%, industrial tariff 0.352 TND/kWh; budget deficit 5-7% GDP, debt ~83% GDP — https://www.trade.gov/country-commercial-guides/tunisia-electrical-power-systems-and-renewable-energy...
28. [Rw11] Agrivoltaics: 20-40% CAPEX premium over utility PV (+$0.30/Wp); NREL range $1.20-2.60/Wp — https://www.mdpi.com/1996-1073/19/1/102...
29. [Rw12] Tunisia land law: foreign ownership of agricultural land prohibited; 40-year leases only; state-domain conversion is an administrative hurdle — https://2009-2017.state.gov/e/eb/ifd/2008/101021.htm...
30. [Rw13] Grid constraints: 3 GW overwhelms local 400 kV; new lines $1-2.5M/km; renewables <5% of generation; curtailment risk — https://reglobal.org/tunisia-focuses-on-grid-expansion-for-integrating-renewable-energy/...
31. [Rc1] Underground DC precedents: Lefdal Mine (olivine; 200 MW; PUE 1.08-1.15 via 8°C fjord water); SubTropolis limestone (18°C + mechanical cooling); no phosphate-mine DC exists — https://www.lefdalmine.com/...
32. [Rc2] Gafsa basin: DInSAR-documented subsidence at Moulares; geothermal gradient 25-30°C/km → 23-29°C at 100-300 m depth — https://www.researchgate.net/publication/251701016_Phosphate_mine_subsidences_deduced_from_differential_interf...

33. [Rc3] Gafsa radiation: ^{226}Ra 16.3-375.1 Bq/kg in soils; phosphate tailings to 7,606 Bq/kg; radon-222 accumulation in enclosed workings; alpha-induced soft errors — https://www.researchgate.net/publication/380540277_Assessment_of_radiation_exposure_in_the_phosphate_mining_ar...
34. [Rc4] ASHRAE TC9.9 Class H1 (5th ed.): 18-22°C recommended, 25°C max for AI accelerators; AI datacenter CAPEX $20-35M/MW (incl. IT), $8-15M/MW facility-only — https://envigilance.com/compliance/ashrae-tc-9-9/...
35. [Rc5] GPU cost: ~100,000 B200-class per 100 MW; $25-40k each → $2.5-4B per 100 MW; 300 MW ≈ $7.5-12B — https://intuitionlabs.ai/articles/data-center-gpu-pricing-2026...
36. [Rc6] US BIS export controls 2024-26: UAE 35,000 GPUs (G42/HUMAIN, 'Compute Diplomacy'); Saudi case-by-case 35,000 cap; Tunisia has no bilateral AI-security agreement — https://www.commerce.gov/news/press-releases/2025/11/statement-uae-and-saudi-chip-exports...
37. [Rc7] Terfezia cultivation: commercial since 1999-2001 (Murcia, Spain); fruiting at year 2-3; 20-yr studies: average ~376 kg/ha, peak 1,069 kg/ha, drought years 0-2 kg/ha — https://pmc.ncbi.nlm.nih.gov/articles/PMC8907101/...
38. [Rc8] Desert-truffle water 200-250 mm/yr; protein 14-22.5% dry weight (the page's '20-27%' sits at the top of the reported range) — https://www.mdpi.com/2073-4395/11/8/1462...
39. [Rc9] Terfezia farm-gate €20-60/kg (€200/kg only in extreme scarcity in Gulf markets); Tuber melanosporum €700-1,500/kg retail — different commodities — https://trufamania.com/desert-truffles.htm...
40. [Rc10] Mycoculture economics: black truffle payback 10-15 years; Terfezia fruits faster but at one tenth the price — https://rovirosafarms.com/melanosporum/...
41. [Rc11] Tunisia sovereign ratings: Moody's Caa1 (Feb 2025 upgrade, stable); Fitch B- (Sep 2025, affirmed Jan 2026); reserves ~4.5 months import cover; WACC 10-15% — https://www.fitchratings.com/research/sovereigns/fitch-upgrades-tunisia-to-b-outlook-stable-12-09-2025...
42. [Rc12] Who finances Tunisia: EIB, AfDB, KfW, World Bank/IFC, EU Global Gateway (17 GW renewable programme); no active IMF program — https://international-partnerships.ec.europa.eu/policies/global-gateway/17-gw-renewable-energy-programme-tunis...
43. [Rc13] Empirical IPP record: 500 MW solar awarded 2024/25 (Scatec, Qair, Voltalia 25-yr PPAs at ~$0.031/kWh); installed renewables ~400 MW; STEG debt >$1.3B — https://www.engineeringnews.co.za/article/scatec-awarded-ppas-for-wind-solar-projects-in-tunisia-2026-01-26...
44. [Rc14] Chain logic: 8 interdependent stages at a generous 70% individual success ⇒ ~5.8% joint probability (0.7^8) — https://www.mdpi.com/2073-4395/14/12/3021...
45. [Rc15] Terfezia claveryi salt sensitivity: establishes only in near-fresh soils, EC 0.82-0.85 dS/m — https://jabonline.in/abstract.php?article_id=1363&sts=2...
46. [Rc16] Terfezia plantation density: ~6,000 mycorrhized plants per hectare — https://www.researchgate.net/publication/299679495_Preparation_and_Maintenance_of_Both_Man-Planted_and_Wild_Pl...
47. [Rc17] Saudi Al-Khalidiah 150-ha Tirmania plantation: largely failed (14 kg/ha after soil import; rainfall collapse) — https://ojs.openagrar.de/index.php/JABFQ/article/download/2184/2570...
48. [Rc18] Desert-truffle market depth: Mamora (Morocco) wild harvest peaks ~100 t/yr; luxury band breaks past ~1,000-1,500 t/yr of new supply — https://www.mdpi.com/1999-4907/14/5/952...
49. [Rc19] Tunisia arid land: ~86,000 ha of state domain earmarked for 40-year agricultural leases (Law 93-120) — https://documents1.worldbank.org/curated/en/926441468778187741/pdf/multi0page.pdf...
50. [Ru1] Uranium from wet-process phosphoric acid: Freeport (US) produced 14M lbs U_3O_8 1978-99 at >95% DEHPA/TOPO recovery; ~20% of US uranium in the 1980s came from phosphates — https://www.osti.gov/servlets/purl/1357599...
51. [Ru2] URANEXT/OCP (Morocco) building the first modern yellowcake-from-phosphoric-acid plant, ~$100M CAPEX, El Jadida, 2025 — https://northafricapost.com/87963-uranext-to-build-moroccos-first-yellowcake-uranium-plant-in-el-jadida.html...
52. [Ru3] WPA uranium SX cost $59-83/lb U_3O_8 (IX $46-76) vs $80-95/lb 2024-26 spot — https://repositories.lib.utexas.edu/bitstreams/89897275-85cf-4cd6-a0c5-1d8daa353799/download...
53. [Ru4] REE from phosphoric acid: PhosAgro pilot; D2EHPA extracts >85% of heavy REEs (Dy, Y, Er) from WPA — https://www.mdpi.com/2673-4117/7/2/58...
54. [Ru5] Gafsa phosphate rock uranium content 45-140 ppm (southern basin) — https://www.researchgate.net/publication/380540277_Assessment_of_radiation_exposure_in_the_phosphate_mining_ar...
55. [H1] Guissouma 2017 (Chemosphere): Tunisia tap-water fluoride 0-2.4 mg/L; 25% of population at dental-fluorosis risk, 20% at skeletal-fluorosis risk (WHO limits) — https://pubmed.ncbi.nlm.nih.gov/28284958/...
56. [H2] Bouchiba 2026 (Toxics 14(5):438): Mdhilla beneficiation effluent Cd 0.49, Pb 0.71, Cr 3.09 mg/L — exceeding discharge limits; risk driven by Cd, Co, Tl — https://doi.org/10.3390/toxics14050438...
57. [H3] Gupta 1996 RCT: fluorosis REVERSED in children (Ca + VitD3 + ascorbic acid; n=25; water 4.5 ppm F; double-blind) — https://pubmed.ncbi.nlm.nih.gov/8942013/...
58. [H4] Ghaemi 2024 (Iran): dental-fluorosis DALY rate up to 981.45 (UI 353.23-1618.40) per 100,000 in a high-F district; Monte Carlo dose-response — https://pubmed.ncbi.nlm.nih.gov/39003868/...
59. [H5] Abtahi 2019 (Water Research): age-sex specific DALYs from drinking-water fluoride, Iran national study — https://pubmed.ncbi.nlm.nih.gov/30953859/...
60. [H6] Naddafi 2022 (Env Research): burden of disease from heavy metals in drinking water, Iran 2019 — https://pubmed.ncbi.nlm.nih.gov/34529973/...
61. [H7] Egner 2014 RCT (n=291): broccoli-sprout beverage 12 wks -> urinary benzene mercapturic acid +61%, acrolein +23% (P<=0.01); GSTT1-positive excrete more — https://pubmed.ncbi.nlm.nih.gov/24913818/...

62. [H8] Pabis 2018: zinc supplementation -> >30% reduction of kidney/liver Cd in aged metallothionein-transgenic mice — https://pubmed.ncbi.nlm.nih.gov/30103140/...
63. [H9] Zhai 2016: Lactobacillus plantarum CCFM8610 binds gut Cd, protects the intestinal barrier, lowers tissue Cd in mice — https://pubmed.ncbi.nlm.nih.gov/27208136/...
64. [H10] Mueller 2018: N95-class respirators >=98% filtration efficiency vs <=44% for any cloth — https://pubmed.ncbi.nlm.nih.gov/29779694/...
65. [H11] Allen & Barn 2020: HEPA purifiers substantially cut indoor PM2.5 and improve subclinical cardiopulmonary health; population benefits often exceed costs — https://pubmed.ncbi.nlm.nih.gov/33241434/...
66. [H12] Drieling 2022 RCT (75 asthmatic children): HEPA in bedroom+living room -> poorly-controlled asthma IRR 0.43-0.45; unplanned clinical utilization IRR 0.35 — https://pubmed.ncbi.nlm.nih.gov/34980119/...
67. [H13] Spirulina RCT (Cambodia, n=179): 2 g/day x 50 days -> anemia reduction significant (p=0.004); weight-gain trend p=0.07 (do not overclaim growth) — https://pubmed.ncbi.nlm.nih.gov/36476193/...
68. [H14] EJAtlas: Gafsa affected population 30,000-70,000; cancer/asthma/infertility reported high; conflict since 2008 — https://ejatlas.org/conflict/phosphate-mining-in-gafsa...
69. [H15] NRGI: 'serious dearth of rigorous studies' linking phosphate operations to health outcomes; 7% birth-defect claim is activist-reported, unverified — https://resourcegovernance.org/articles/uncovering-impacts-phosphate-mining-tunisian-women...
70. [H16] Geographical 2024: asthma onset age 2 documented (Redeyef); nearest hospital 70 km; doctors prescribe leaving Gafsa — https://geographical.co.uk/science-environment/gafsas-silent-suffering-the-hidden-cost-of-phosphate-mining-in-...